higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

MATHEMATICAL LITERACY

(Second Paper)

NQF LEVEL 4

(10401034)

25 February 2019 (Y-Paper)
13:00–16:00

Non-programmable calculators and mathematical instruments may be used.

This question paper consists of 11 pages and 2 addenda.

NCV

TIME: 3 HOURS
MARKS: 150

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Clearly show ALL calculations.
5. Round off answers to TWO decimal places, unless stated otherwise.
6. Use $\pi = 3,14$. Students will be penalised if any other value is used.
7. Diagrams are NOT necessarily drawn to scale.
8. Answer QUESTION 2.4 on ADDENDUM A and hand it in with the ANSWER BOOK.
9. Answer QUESTION 3.1.4 on ADDENDUM B and hand it in with the ANSWER BOOK.
10. Write neatly and legibly.

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QUESTION 1

1.1 Sally has a business that plans large parties for schools or companies. Given below is her budget and income statements for the last two parties she hosted.
Both parties expected 200 guests.

Study the budget and answer the questions.

	PARTY 1		PARTY 2	
	Projected	**Actual**	**Projected**	**Actual**
INCOME				
Fees paid	50 000	46 250	64 000	67 840
Donations	4 000	4 000	4 000	3 600
Total income:	**54 000**	**50 250**	**68 000**	**71 440**
EXPENSES				
Prizes/Gifts	4 000	5 000	5 000	5 200
Catering	24 000	22 200	32 000	33 920
Labour	2 000	2 000	2 000	1 800
Venue hire	6 000	6 000	6 000	6 000
Decorations	8 000	7 500	7 500	8 200
Total expenses:	**44 000**	**42 700**	**52 500**	**55 120**
Profit/Loss:	**10 000**	**7 550**	**15 500**	**16 320**

1.1.1 Calculate the cost per person for catering for PARTY 1 and PARTY 2. Use the projected amounts for each calculation. (2)

1.1.2 Give a possible reason as to why the catering cost is different for each party. (2)

1.1.3 What does the term *variance* mean in the financial world? (2)

1.1.4 (a) What is the variance between the projected and actual amounts of the profit/loss for each party? (2)

(b) What does the variance between the projected and actual amounts of the profit/loss for each party indicate to Sally? (2)

1.1.5 What advice would you give Sally when she plans for future parties? (2)

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1.2 The following table shows a comparison between water and electricity tariffs in Sally's municipality over two years.

Study the table and answer the questions.

Service	**Units/Amount**	**2016/17 tariff includes VAT per kℓ**	**Units/Amount**	**2017/18 tariff includes VAT per kℓ**
Water **Domestic**	Per month		Per month	
Step 1	Less than 6 kℓ	R0,00	0 – 6 kℓ	R4,56
Step 2	6 – 10,5 kℓ	R15,68	6 – 10,5 kℓ	R17,75
Step 3	10,5 – 20 kℓ	R20,02	10,5 – 20 kℓ	R24,93
Step 4	20 – 35 kℓ	R32,65	20 – 35 kℓ	R41,53
Step 5	35– 50 kℓ	R49,83	35 – 50 kℓ	R70,29
Step 6	Above 50 kℓ	R93,39	Above 50 kℓ	R238,59
Electricity	Per month	**2016/17 tariff excludes VAT per kWh**		**2017/18 tariff excludes VAT per kWh**
Domestic	Block 1: 0 – 600 kWhs	R1,87	Block 1: 0 – 600 kWhs	R1,92
	Block 2: 600 kWhs +	R2,28	Block 2: 600 kWhs +	R2,34

[Source: *www.greenaudits.co.za*]

1.2.1 Sally used 16 kℓ of water and 400 kWh of electricity in January 2018.

Show that the total cost including VAT is R1 119,88. (7)

1.2.2 Why do you think the tariff rates of water and electricity increase as the amount used increases? (2)

NCV

1.3 Sally decides to take out a loan to buy some equipment. The bank agrees to give her a loan of R10 000 at an interest rate of 13,8%. The loan is to be repaid over a period of two years. The table below shows the amounts to be paid over the two-year period.

$$\text{Interest} = \frac{13{,}8\% \text{ of amount owing}}{12}$$

Month	Opening balance	Interest	Total	Payment	Closing balance
1	R10 000,00	R115,00	R10 115,00	R490,00	R9 625,00
2	R9 625,00	R110,69	R9 735,69	R490,00	R9 245,69
3	R9 245,69	R106,33	R9 352,01	R490,00	R8 862,01
4	R8 862,01	R101,91	R8 963,93	R490,00	**R8 473,92**
5	R8 473,93	R97,45	R8 571,38	R490,00	R8 081,38
6	R8 081,38	R92,94	R8 174,31	R490,00	R7 684,31
7	R7 684,31	R88,37	R7 772,68	R490,00	R7 282,68
8	R7 282,68	R83,75	R7 366,43	R490,00	R6 876,43
9	R6 876,43	R79,08	R6 955,51	R490,00	R6 465,51
10	R6 465,51	R74,35	R6 539,86	R490,00	R6 049,86
11	R6 049,86	R69,57	R6 119,44	R490,00	R5 629,44
12	R5 629,44	R64,74	R5 694,18	R490,00	R5 204,18
13	R5 204,18	R59,85	R5 264,02	R490,00	R4 774,02
14	R4 774,02	R54,90	R4 828,93	R490,00	R4 338,93
15	R4 338,93	R49,90	R4 388,82	R490,00	R3 898,82
16	R3 898,82	R44,84	R3 943,66	R490,00	R3 453,66
17	R3 453,66	R39,72	R3 493,38	R490,00	R3 003,38
18	R3 003,38	R34,54	R3 037,92	R490,00	R2 547,92
19	R2 547,92	R29,30	R2 577,22	R490,00	R2 087,22
20	R2 087,22	R24,00	R2 111,22	R490,00	R1 621,22
21	R1 621,22	R18,64	R1 639,86	R490,00	R1 149,86
22	R1 149,86	R13,22	R1 163,09	R490,00	R673,09
23	R673,09	R7,74	R680,83	R490,00	R190,83
24	R190,83	R2,19			R0,00

1.3.1 Show how the closing balance of R8 473,92 in the fourth month was calculated. (4)

1.3.2 How much would the last payment in the 24th month be? (2)

1.3.3 How much does Sally actually pay over the two-year period? (2)

1.3.4 What do you think would happen if she paid R560 per month? (2)

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1.4 Use the table below to answer the questions.

RATES OF TAX FOR INDIVIDUALS

2018 tax year (1 March 2017 – 28 February 2018)

Taxable income (R)	Rates of tax (R)
0 – 189 880	18% of taxable income
189 881 – 296 540	34 178 + 26% of taxable income above 189 880
296 541 – 410 460	61 910 + 31% of taxable income above 296 540
410 461 – 555 600	97 225 + 36% of taxable income above 410 460
555 601 – 708 310	149 475 + 39% of taxable income above 555 600
708 311 – 1 500 000	209 032 + 41% of taxable income above 708 310
1 500 001 and above	533 625 + 45% of taxable income above 1 500 000

Tax rebates

Tax rebate	Tax year				
	2018	**2017**	**2016**	**2015**	**2014**
Primary	R13 635	R13 500	R13 257	R12 726	R12 080
Secondary (65 and older)	R7 479	R7 407	R7 407	R7 110	R6 750
Tertiary (75 and older)	R2 493	R2 466	R2 466	R2 367	R2 250

[Source: *www.sars.gov.za*]

1.4.1 Sally's brother has asked her to calculate his monthly tax amount for 2018. He earns an annual taxable income of R400 000 and is 45 years old. Sally calculates that he will have to pay R6 695,63 per month.

Is Sally's calculation correct. Verify, by showing ALL calculations. (7)

1.4.2 If he gets a salary increase of R20 000 per annum what will happen to the amount that he would have to pay for tax? Give a reason for your answer. (2)

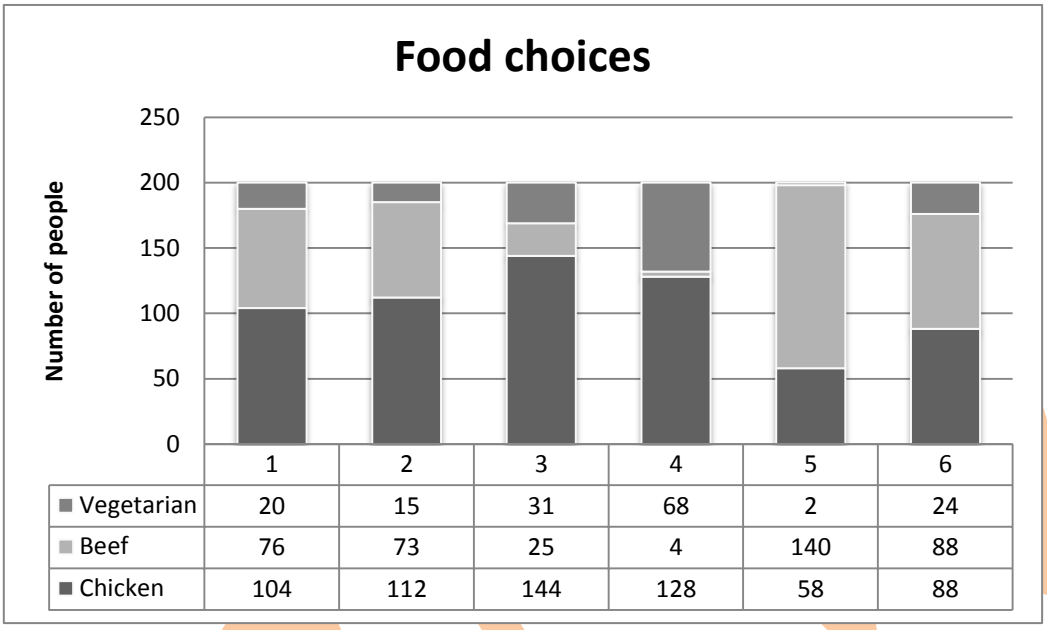
[40]

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QUESTION 2

2.1 Sally kept a record of the most popular food choices made by the guests at six parties and drew up the graph below. Each party catered for 200 guests.

Study the graph and answer the questions.



	1	2	3	4	5	6
Vegetarian	20	15	31	68	2	24
Beef	76	73	25	4	140	88
Chicken	104	112	144	128	58	88

2.1.1 What measure of central tendency would be the most useful for Sally to consider? Give a reason for the answer. (2)

2.1.2 What is the sample size of this survey in terms of people? (1)

2.1.3 What other type of graphs could be drawn to show which food choice is more popular? Substantiate the answer. (2)

2.1.4 Compare the averages for the number of guests choosing chicken to those that are choosing beef. (5)

2.1.5 Explain the term *probability*. (2)

2.1.6 If a person is chosen at random from the first party, what is the probability that he/she would choose chicken? Give the answer as a simplified fraction. (2)

2.1.7 If a person is chosen at random from ALL the parties, what is the probability that he/she would choose beef? Round off the answer to the nearest two-decimal fraction. (3)

2.1.8 If a person is chosen at random from the first THREE parties, what is the probability that he/she would choose chicken or vegetarian? Round off the answer to the nearest percentage. (4)

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2.2 When collecting data, it is important that the sample is unbiased.

2.2.1 Explain what the term *"unbiased"* means in relation to the above statement. (2)

2.2.2 Discuss TWO ways to ensure that the sample is unbiased. (2 × 2) (4)

2.3 Give ONE advantage and ONE disadvantage for each of the following data-collection methods:

2.3.1 Observation

2.3.2 Questionnaire

(2 × 2) (4)

2.4 Sally wants to compare projected and actual costs across different cost items.

Draw a compound vertical bar graph on ADDENDUM A (attached) to illustrate the information in the following table:

COST ITEM	PROJECTED	ACTUAL
Prizes/Gifts	R4 000	R5 000
Catering	R24 000	R22 000
Venue hire and decorations	R14 000	R13 500

(6)
[37]

QUESTION 3

3.1 Sally decides to offer clients two pricing options for a function.

Option A: R6 500 plus R250 per person
Option B: R300 per person

The following table shows the total cost of the function for each pricing option:

Number of people	0	75	100	150	175	**C**
Option A	6 500	25 250	**A**	44 000	50 250	56 500
Option B	0	22 500	30 000	45 000	**B**	60 000

3.1.1 Calculate the missing values of A, B and C as shown in the table. (3 × 2) (6)

3.1.2 Derive a formula for Option A. (3)

3.1.3 Derive a formula for Option B. (2)

3.1.4 Draw TWO graphs on ADDENDUM B (attached) showing each cost option. Clearly indicate the break-even point. (7)

3.1.5 What advice would you give potential clients about which option they should choose? (4)

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3.2 Sally wants to work out how many staff members she needs to set up the venue. She draws up the following table:

Number of people	1	2	3	4	6	**B**
Number of hours	12	**A**	4	3	2	1

3.2.1 Calculate the value of A and B by using the following formula:

$$\text{Number of hours} = \frac{12}{\text{Number of people}}$$ (4)

3.2.2 What type of relationship is represented in this table? Give a reason for the answer. (3)

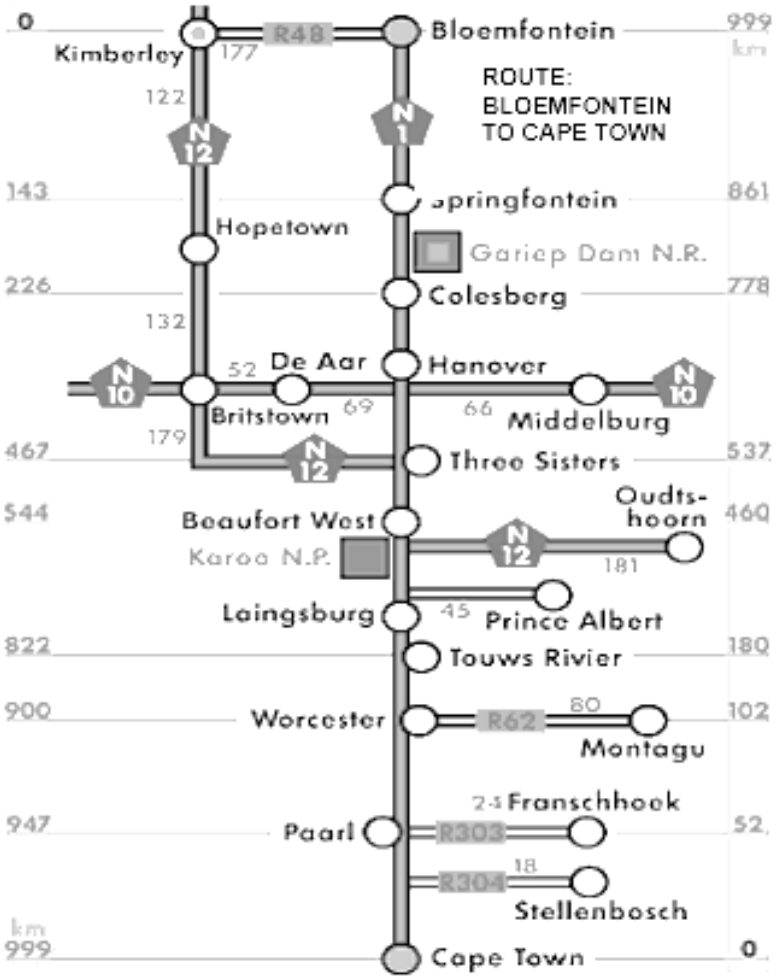
3.2.3 Give TWO factors that Sally should consider when making her decision. (4)

[33]

QUESTION 4

4.1 The map below shows the distance between Bloemfontein and Cape Town.

Study the map and answer the questions.



[Source: *www.southafrica-travel.net*]

4.1.1 What is the distance between Bloemfontein and Cape Town? (2)

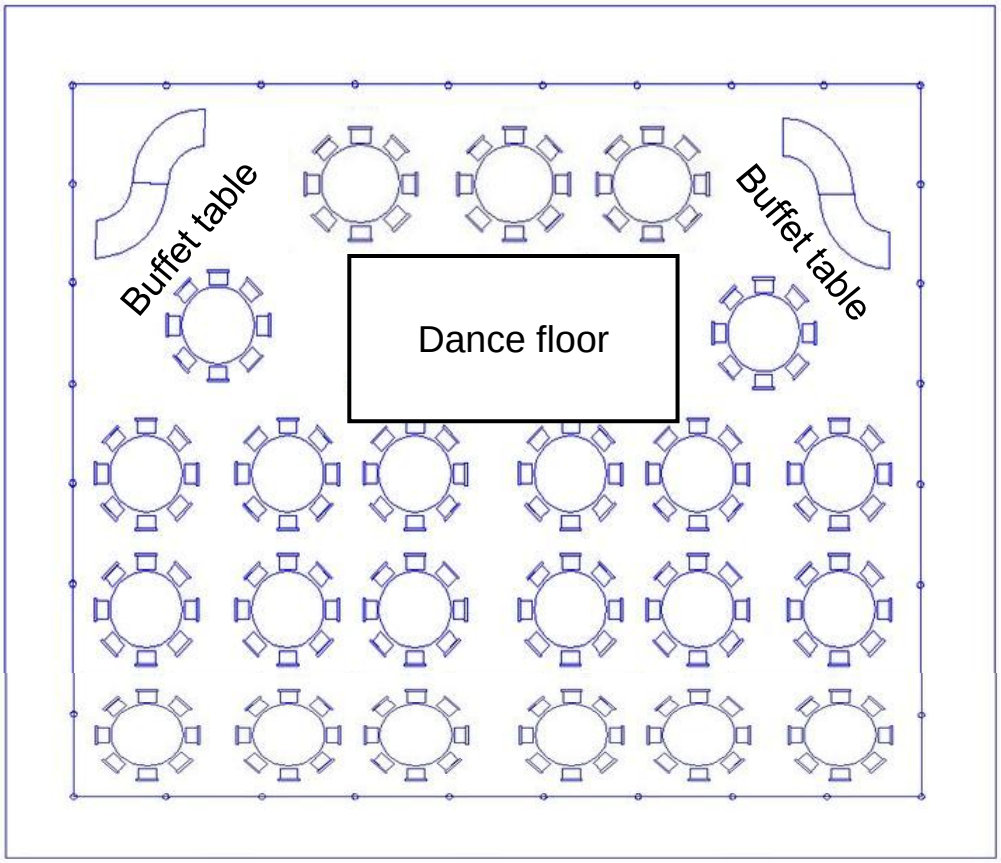
4.1.2 On which national roads would one travel from Kimberley to Oudtshoorn? (2)

4.1.3 What is the distance from Cape Town to Montagu? (3)

4.1.4 Show, by calculation, that the distance from Kimberley to Middelburg via Britstown is 441 km. (3)

4.2 The floor plan below shows one of the venues that Sally can use for her functions. Study the plan and answer the questions.

NOTE: The plan is NOT drawn to scale.



4.2.1 If each circular diagram represents a table and chairs around the table, show by a calculation that the venue can accommodate more than 180 people. (3)

4.2.2 The scale on the plan is 1:200. Each set of table and chairs needs a diameter of 3 m.

Calculate the size (in mm) of each set of table and chairs on the plan. (3)

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4.2.3 Sally is making square overlays for each table (an overlay is a cloth that gets placed in the middle of the table over the normal tablecloth). The side of each overlay will be 80 cm.

Calculate how many square metres of fabric Sally will need in total for all the tables.

Area of square = side × side (5)

4.2.4 Sally would also like to stitch a satin ribbon around the edge of each overlay.

If satin ribbon costs R3,20 per metre, calculate the cost of stitching a satin ribbon around the edge of each overlay.

Perimeter of square = side × 4 (7)

4.3 Sally is considering using a polystyrene cooler box (shown below) to transport some of the food for the function.

Use the given information to answer the questions.



Internal dimensions:

Length: 45 cm
Breadth: 26 cm
Height: 26,5 cm

External dimensions:

Length: 51,5 cm
Breadth: 32,5 cm
Height: 33 cm

Volume = l × b × h
1 litre = 1 000 cm^3

4.3.1 How many litres can the polystyrene cooler box hold? Show ALL calculations. (5)

4.3.2 Sally needs to transport 80 litres of fruit salad in these boxes. She finds the following boxes:

22 litre box for R350
30 litre box for R420

(a) Calculate how many boxes of each size he would need. (4)

(b) Calculate the total cost of transporting each type of box. (2)

(c) Which would be the most cost-effective option? (1)

[40]

TOTAL: 150

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ADDENDUM A

CENTRE NUMBER: | | | | | | | | | | | |

EXAMINATION NUMBER: | | | | | | | | | | | | |

Hand in with the ANSWER BOOK.

QUESTION 2.4



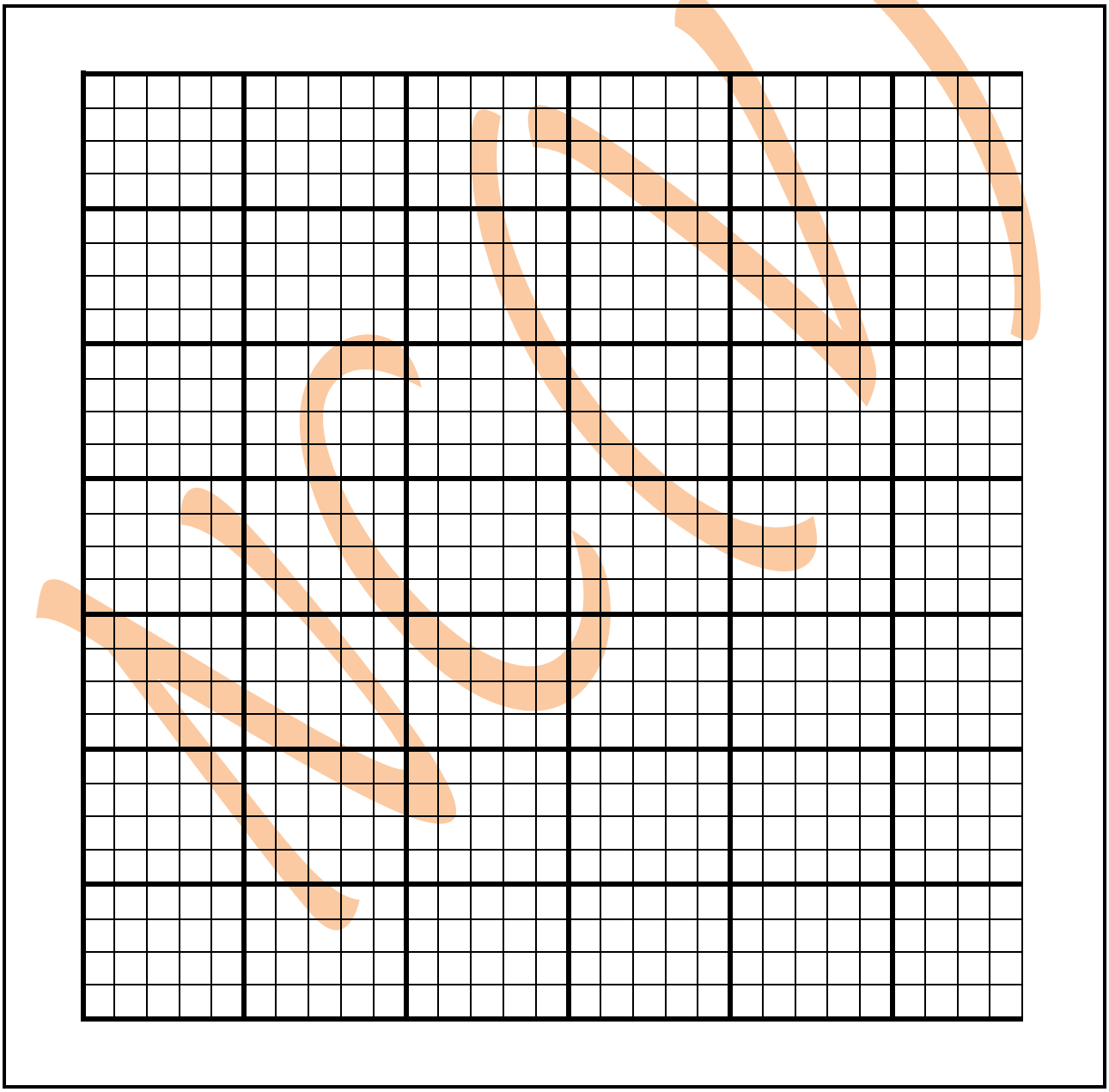
ADDENDUM B

CENTRE NUMBER:

EXAMINATION NUMBER:

Hand in with the ANSWER BOOK.

QUESTION 3.1.4



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